20 There are 9 counters in a bag.

There is an even number on 3 of the counters.
There is an odd number on 6 of the counters.

Three counters are going to be taken at random from the bag.
The numbers on the counters will be added together to give the total.

Find the probability that the total is an odd number.

Odd total:one odd and two even, or three odd.

1 mark
Denominators9,8,7

OOO $\frac{6}{9} \times \frac{5}{8} \times \frac{4}{7} = \frac{120}{504}$

1 mark
One correct product

OEE $\frac{6}{9} \times \frac{3}{8} \times \frac{2}{7} = \frac{36}{504}$

EOE $\frac{3}{9} \times \frac{6}{8} \times \frac{2}{7} = \frac{36}{504}$

1 mark
All four products

EEO $\frac{3}{9} \times \frac{2}{8} \times \frac{6}{7} = \frac{36}{504}$

P(odd) = (120 + 36 + 36 + 36) ÷ 504

1 mark
Add disjoint cases

= 228/504 = 19/42

1 mark
19/42

(Total for Question 20 is 5 marks)